



BRIGMAN

Deep Sea Technology

1600 Pennsylvania Ave NW
Washington DC
20500, United State

Requirements Analysis Report (BCIC)

ICT2113 Software Modelling and Analysis — Lecture 5

Vendor: **Brigman Deep Sea Technology**

Team Name: **TEAM_NAME**

NAME_1 (ID_1)

NAME_2 (ID_2)

NAME_3 (ID_3)

NAME_4 (ID_4)

NAME_5 (ID_5)

SUBMISSION_DATE





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1 Purpose of This Document

This document records the requirements analysis performed on the Project Specification (PS) using the BCIC method (Breakdown, Clarify, Interpret, Categorize). The goal is to conduct regulated-revision: identify incongruities, clarify intent, revise requirements into clear/testable statements, and organize them into a consistent structure.

2 Inputs and Scope

- Analysed Inputs: PS vVERSION, tender notes, requirements tree, selected models.
- In-scope: requirement clarity, testability, consistency, completeness gaps.
- Out-of-scope: proposing implementation/design solutions beyond requirement intent.

3 Method Used: BCIC

- **Breakdown:** Identify incongruent requirements (ambiguous, non-testable, missing info, contradictions, compound).
- **Clarify:** Determine the intended meaning/scope using stakeholder info, PS context, assumptions, and notes.
- **Interpret:** Rewrite into clear, atomic, testable requirements using imperative wording (shall/must).
- **Categorize:** Place revised requirements into the correct section/level (Biz Ops vs IT, L0/L1/L2, precedence).





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4 Breakdown Findings (Issue List)

Issue ID	PS Ref	Problem Type	What is wrong (short)
BR-01	BO-L1-?	Ambiguous	e.g., “monitor” not defined: frequency and scope unclear.
BR-02	IT-L1-?	Not testable	e.g., “secure” without measurable criteria.

5 BCIC Worked Example (One Full Item)

Example Item: ISSUE_ID_OR_REQ_ID

Original PS Statement (as written):

“The system should monitor vessel health.”

Breakdown (what is wrong):

- Ambiguous term: “monitor” (continuous vs periodic?).
- Scope unclear: what counts as “vessel health”?
- Not fully testable without specifying what must be monitored.

Clarify (intended meaning / assumptions used):

- Monitoring is expected during mission execution to support safety-first operations.
- Vessel health refers to critical subsystems such as hull integrity, propulsion, and power systems.

Interpret (revised requirement(s), clearer + testable):

- **R-NEW-01:** The system **shall continuously monitor** vessel health during mission execution.





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Requirements Analysis (BCIC)

- **R-NEW-02:** Vessel health monitoring **shall include** hull integrity, propulsion status, and power system status.

Categorize (where it belongs and why):

- Category: IT / Technical (because it is a system capability).
- Level: L1 (capability) + L2 (details of what is included).
- Precedence: Essential (supports safety and mission continuity).

6 Revised Requirements Summary

- Total issues identified: N
- Total requirements revised: N
- Total new atomic requirements created (from splitting): N

7 Conclusion

The BCIC analysing method was applied to perform regulated-revision of the PS. Incongruities were identified, clarified, revised into testable statements, and categorized into the appropriate structure. The revised requirements improve clarity and consistency while remaining faithful to the original project intent, and are ready for subsequent specification work.

